

AMATH 732

Quiz 4; 30 Minutes

Question 1: 7 marks Consider the nonlinear weakly damped oscillator problem from the notes with a different IC.

$$\frac{d^2x}{dt^2} + \epsilon \frac{dx}{dt} + x = 0$$

with the initial conditions $x(0) = 2$, $v(0) = 0$. Use the method of multiple scales to find the approximate solution.

Question 1 : more space

Question 2: 3 marks A set of masses on springs is connected in series, attached to a wall, and placed into a box that you cannot see into, or otherwise measure inside. One mass sticks out of the box and you can force it as you would like. Assuming all damping is weak so that all motion is underdamped explain how you would determine how many masses are inside the box. You do not need to do calculations, but explain clearly.